

BEYOND MEAN FOILS: AUDITING WORST-FOIL SPECIFICITY IN FROZEN CLIP REGION EXPLANATIONS

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ABSTRACT

Spatial overlap and mean class contrast need not imply target-specific region responses. We audit CCI’s maximum-target-drop region in frozen CLIP, using separate prompts for selection and evaluation. Across COCO and VOC with two checkpoints, 41.16–64.78% of locality/mean-foil screen-passers fail a worst-foil criterion. Fixing regions and screening populations, excluding annotated-present foils leaves 39.77–63.64% failing. Exclusion reduces failure beyond exact cardinality-matched random removal by 0.86–0.92 percentage points on COCO; VOC intervals include zero. Positive-target filtering retains high failure. Among screened failures, unrestricted eight-region oracles permit sign repair in 6.25–7.84% on COCO and 27.40–31.29% on VOC. A target-drop tolerance of $\epsilon = .02$ reduces feasible repair to 0.16–0.98%; relaxing it increases opportunity and exposes target and locality costs.

Index Terms— CLIP, explanation auditing, class specificity, constrained region selection

1 INTRODUCTION

CLIP [1] region explanations often use intervention-induced image–text score changes. Cluster-based Concept Importance (CCI) [2] combines cluster importance into a heatmap; we audit only its maximum-target-drop region, CCI-top1. Target contribution, spatial overlap, and relative specificity differ: one region may support several genuinely present classes.

A positive mean contrast need not imply a positive worst-foil margin. We quantify this gap, test whether annotated co-occurrence explains it, and ask whether another candidate can resolve it under a target-response budget. COCOA [3] defines corpus–foil attribution; CASE [4] diagnoses class-insensitive saliency and builds contrastive maps. Related CLIP attribution [5, 6, 7], class/concept contrasts [8, 9, 10], and caption foils [11] motivate this audit. Our contribution is fixed-population foil control and exact decomposition of candidate, budget, and selector bottlenecks for an already selected re-

gion. Worst-foil (WF) reranking is a diagnostic probe. Unlike IDEA’s few-shot adaptation [12], the model stays fixed.

2 AUDIT AND REPAIR OPPORTUNITY

For checkpoint m , the pre-softmax score is $s_m(x, c) = \tau_m \langle z_m(x), e_m(c) \rangle$, with unit-norm embeddings and learned $\tau_m = \exp(\ell_m)$. Region R has intervention response

$$d_R(c) = s_m(x, c) - s_m(\mathcal{I}(x, R), c). \quad (1)$$

For target category t , selection uses “a photo of a {category}”:

$$R_{\text{CCI}} = \arg \max_{R \in \mathcal{R}(x)} d_R^{\text{sel}}(t). \quad (2)$$

Held-out margins and screens. Evaluation uses “a picture of the {category}”, “an image containing a {category}”, and “the {category}”. For unit-norm text embeddings $e_h(c)$, let $H = 3$, $n_c = \|H^{-1} \sum_h e_h(c)\|_2$, and $\bar{d}_c = H^{-1} \sum_h d_h(c)/n_c$. With $\mathcal{F}_t$ the non-target dataset categories, the aggregate worst-foil margin is

$$P(R, t) = \bar{d}_t - \max_{f \in \mathcal{F}_t} \bar{d}_f. \quad (3)$$

This equals using normalized mean text embeddings; class-specific n_c can change rankings. Tables report raw target drop $H^{-1} \sum_h d_h(t)$, whereas the positive-target screen uses $\bar{d}_t$. Evaluation templates are not used to select a region.

For the same CCI-top1 region, define $A = \{\text{bbox} \geq .5, \bar{d}_t - \text{mean}_f \bar{d}_f > 0\}$ and $F = \{P < 0\}$. The diagnostic screen and .5 threshold are study-defined, not a standard of explanation validity. Because a positive mean-foil margin does not require positive target drop, we also audit $A^+ = A \cap \{\bar{d}_t > 0\}$. The primary repair population is $B = A \cap F$, fixed before reranking. Negative P is relative to the declared foil ontology, not a judgment of semantic invalidity.

Target-constrained diagnostic probe. Define

$$M^{\text{sel}}(R, t) = d_R^{\text{sel}}(t) - \max_{f \in \mathcal{F}_t} d_R^{\text{sel}}(f), \quad (4)$$

$$\mathcal{R}_\epsilon = \{R : d_R^{\text{sel}}(t) \geq d_{R_{\text{CCI}}}^{\text{sel}}(t) - \epsilon\}, \quad (5)$$

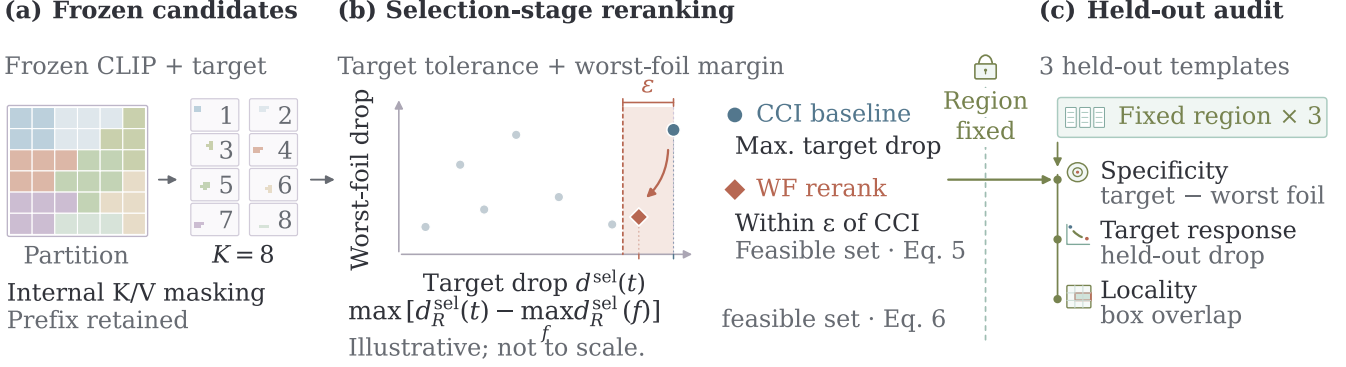


Figure 1: Selection and held-out audit. A region is fixed using selection-stage responses before three-template evaluation of specificity, target response, and locality. The candidate-response diagram is illustrative, not measured data.

and worst-foil (WF) reranking

$$R_{\text{WF}} = \arg \max_{R \in \mathcal{R}_\epsilon} M^{\text{sel}}(R, t). \quad (6)$$

The baseline is feasible, so any switch loses at most ϵ of *selection-stage* target drop; this gives no held-out guarantee. Mean and Max-1 instead maximize $d_t^{\text{sel}} - \text{mean}_f d_f^{\text{sel}}$ and $d_t^{\text{sel}} - .1 \max_f d_f^{\text{sel}}$ over the same set. The reference budget is $\epsilon = .02$ in logit units, not 2% relative loss. A post-hoc .01/.02/.05/.10/.20 sweep selects no new optimum. Cosine budgets ϵ/τ_m are not calibrated across checkpoints; unrestricted oracles remove the budget entirely.

Exact diagnostic oracles. For a baseline-failure population B , the feasible repair count is

$$N_O(B, \epsilon) = \sum_{i \in B} \mathbf{1} \left\{ \max_{R \in \mathcal{R}_\epsilon(i)} P_i(R) \geq 0 \right\}. \quad (7)$$

The unrestricted oracle searches all K candidates. A sign pass means $P_i(R) \geq 0$, without locality or positive-target constraints. Failures split into: no passing candidate (C_0); passing candidates but *none feasible* (C_1); a feasible pass missed by WF (C_2); and WF sign repair (r). Thus $|B| = C_0 + C_1 + C_2 + r$ and $N_O = C_2 + r$. We additionally count repairs satisfying $J(R) = \{P_i(R) \geq 0, A(R), \bar{d}_t(R) > 0\}$; $A(R)$ applies the same full-foil screen to the new region, without changing original B . Oracles use held-out data only for diagnosis.

Fixed-population foil controls. For image i , let $\mathcal{F}_i^{\text{abs}}$ remove all annotated-present categories from the non-target foils. Hold R_{CCI} and the full-foil screen A fixed; $\mathcal{F}_i^{\text{abs}}$ tests $\bar{d}_t - \max_{f \in \mathcal{F}_i^{\text{abs}}} \bar{d}_f < 0$. No region or screen is reselected. Reducing the foil set cannot increase failure, so we also match its cardinality. For N_i full foils, $k_i = |\mathcal{F}_i^{\text{abs}}|$, and m_i foils exceeding the target, a uniform size- k_i subset without replacement fails with probability

$$q_i = 1 - \frac{\binom{N_i - m_i}{k_i}}{\binom{N_i}{k_i}}. \quad (8)$$

With zero numerator when $k_i > N_i - m_i$, we compare $\Delta_A = |A|^{-1} \sum_{i \in A} (\mathbf{1}\{\mathcal{F}_i^{\text{abs}}\} - q_i)$. Restricted sets are nonempty. Annotation absence does not establish semantic absence; crowd annotations remain excluded.

3 PROTOCOL AND REPRODUCIBILITY

COCO val2014 [13] and VOC2007 test [14] provide 27,708/1,943 eligible images, each evaluated with OpenAI CLIP B/16 and B/32 (59,302 image-model records). The target is the largest instance’s category, using annotation area on COCO and box area on VOC. We require another category and 2–80% target area; COCO crowd annotations are excluded. Full foils are the other 79/19 classes.

Projected patch features from the visual transformer are L2-normalized. K-means uses $K = 8$, three initializations, 50 iterations, and seed $1701 + i$; sample index i follows sorted image IDs shuffled with seed 1701. Intervention $\mathcal{I}$ adds $-\infty$ to attention logits at selected patch-key columns before softmax in every block/head, retaining the class-token column. Ties use first argmax; PyTorch 2.8.0 MPS features are cast to float32 before normalization. BBox precision is the fraction of selected patch centers in target-class box unions, mapped by $224/\min(W, H)$ and center-crop offsets; it measures neither pixel precision nor target coverage [15].

Float32 candidate analyses use stored responses. Foil controls regenerate full-class responses: IDs, CCI labels, feasible sizes, and full-foil states match; compared summaries differ by less than 10^{-5} . Original A, A^+ memberships are retained. Missing original masks prevent an identity check.

Intervals use 10,000 image bootstraps and percentile 2.5/97.5 limits, pointwise without multiplicity correction. Paired quantities are differenced within image before re-sampling. Foil comparisons resample within fixed A/A^+ ; repair-cost intervals condition on actual repairs. Exact q_i is primary; 100 seeded draws check it.

4 RESULTS

Table 1: Fixed original A : full-foil, annotation-absent, and exact cardinality-matched random failure rates (%). Δ_A is absent minus random in percentage points, with paired 95% CI.

Setting	Full	Absent	Random	Δ_A
COCO B/16	64.37	63.17	64.09	-0.92 [-1.08,-0.77]
COCO B/32	64.78	63.64	64.50	-0.86 [-1.01,-0.72]
VOC B/16	42.27	40.94	41.46	-0.52 [-1.16,+0.06]
VOC B/32	41.16	39.77	40.32	-0.55 [-1.19,+0.04]

Failure persists after annotation exclusion. Screen sizes are 17,726, 17,847, 1,209, and 1,227 in table order. Exclusion resolves only 213, 203, 16, and 17 full-foil failures; 96.63–98.24% remain (failure counts in Table 2). The rate falls by 1.20/1.14/1.32/1.39 percentage points. Table 1 separates this change from foil-count reduction: COCO exclusion has a larger effect than random removal of the same number of categories, whereas VOC Δ_A intervals include zero. High failure is therefore not explained solely by explicitly annotated co-occurring classes under this protocol.

Positive-target and scene-count checks. Requiring positive target drop leaves 16,855/17,330/1,186/1,215 images. Full-foil failure in A^+ is 62.64/63.79/41.40/40.74%; annotation-absent failure remains 61.38/62.61/40.05/39.34%. Matched-random comparisons retain the COCO/VOC interval pattern. Even scenes with exactly two annotated categories have COCO annotation-absent failure of 63.39/64.95%.

Table 2: Exact local decomposition of *screened failures* $B = A \cap F$ at $\epsilon = .02$. Counts in C_0, C_1, C_2, r are mutually exclusive; $C_2 + r$ is the feasible-oracle count.

Setting	$ B $	C_0	C_1	C_2	r
COCO B/16	11,410	10,697	695	0	18
COCO B/32	11,561	10,655	875	1	30
VOC B/16	511	371	135	0	5
VOC B/32	505	347	155	1	2

Candidate and budget bottlenecks differ by dataset. No region among the eight candidates passes for 93.75/92.16% of COCO screened failures, versus 72.60/68.71% on VOC (Table 2). Removing the budget would therefore permit sign repair in only 6.25/7.84% of COCO B , versus 27.40/31.29% on VOC. Figure 2 summarizes capacity within these eight candidates. The .02 budget then excludes 96.43–98.10% of those unrestricted opportunities: for COCO B/16, 695 of 713 are excluded, leaving 18. Across settings, feasible-oracle counts are 18/31/5/3 (0.16/0.27/0.98/0.59% of B), and WF finds 18/30/5/2.

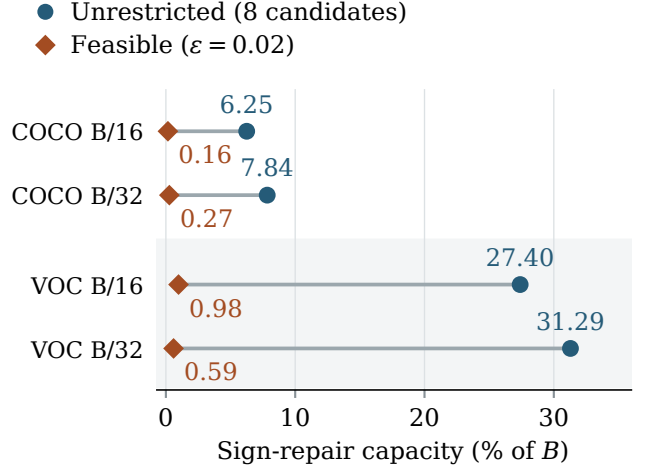


Figure 2: Budget-constrained sign-repair capacity among fixed screened failures B . Connected markers compare the same setting on a shared linear scale. Unrestricted: at least one of all eight candidates has $P \geq 0$; feasible: at least one also meets $\epsilon = .02$. Percentages use each setting’s own $|B|$ (Table 2); these are exact counts, not margin gains or confidence intervals.

Table 3: Local COCO B/16 tolerance sweep. Multi is the all-image percentage with $|\mathcal{R}_\epsilon| > 1$; O_B, r_B are oracle/actual repair percentages of *fixed* B . Target and bbox columns show *global impacts*: all-image WF–CCI means, not costs conditional on repair. Bbox is in percentage points.

ϵ	Multi	O_B	r_B	Δd_t	100 Δ bbox
0.01	2.46	0.07	0.07	+0.00013	-0.049
0.02	4.71	0.16	0.16	+0.00005	-0.118
0.05	11.48	0.42	0.41	-0.00050	-0.229
0.10	20.76	0.78	0.74	-0.00205	-0.546
0.20	35.78	1.57	1.50	-0.01397	-1.626

More opportunity requires a looser budget. In Table 3, increasing ϵ from .02 to .20 raises the multi-candidate fraction from 4.71% to 35.78% and screened-failure oracle repair from 0.16% to 1.57%. Mean selection-target loss rises from .000245 to .023805. At .20, held-out raw target change is $-.01397$ [$-.01520, -.01275$], and bbox change is -1.63 points [$-1.88, -1.37$]. At .20, O_B is 2.05/6.65/7.13% for COCO B/32 and VOC B/16/B/32. At .02, 95.29% of COCO B/16 images have just one feasible candidate. Thus the low repair rate at that budget must be read alongside the unrestricted oracle in Table 2, not as budget-independent repair capacity.

Global impact is not the cost of repair. Table 3 averages costs over all images, including unchanged regions. Within B , the 254/270/16/7 switches at .02 instead have mean bbox changes of $-28.80/-24.92/-39.60/-35.52$ points. Table 5 is narrower still: it conditions on actual sign repairs in B . Conditional costs are neither all-image nor causal effects.

Table 4: Mean aggregate-margin differences over all eligible images, WF minus each alternative, with paired 95% CIs. Values are multiplied by 10^3 ; D counts WF/Max-.1 selection disagreements at $\epsilon = .02$.

Setting	WF-Mean	WF-Max-.1	D
COCO B/16	12.461	0.245	84
	[10.879,14.138]	[0.128,0.374]	
COCO B/32	9.343	0.241	89
	[7.986,10.801]	[0.083,0.414]	
VOC B/16	4.480	0.507	5
	[2.318,7.054]	[0.031,1.111]	
VOC B/32	3.184	0.418	5
	[1.467,5.133]	[0.058,0.927]	

Small baseline gains arise from few disagreements. WF and Max-.1 differ on only 84/89/5/5 images (Table 4). Their agreement is over 99.6% overall but 93.56/92.70/94.62/94.57% within multi-candidate sets. On the 84 COCO B/16 disagreements, WF gains .0807 margin [.0443, .1171], while raw target drop changes by $-.0326$ [$-.0628$, $-.0033$]. The whole-sample gain is .000245. The VOC disagreement subsets contain only five images each.

Table 5: Actual sign repairs *within original B*, $\epsilon = .02$. r counts repairs; j counts those also satisfying J . WF-CCI means with repair-subset 95% CIs. Δd_t is raw held-out target-score change; only bbox change is in percentage points.

Setting	r	j	Δd_t	100 Δ bbox
COCO B/16	18	12	0.029	-19.53
			[-0.095,0.148]	[-38.61,-1.17]
COCO B/32	30	25	0.094	-14.04
			[0.037,0.149]	[-27.61,-1.48]
VOC B/16	5	3	-0.021	-16.87
			[-0.170,0.182]	[-26.11,-7.00]
VOC B/32	2	1	-0.078	-28.17
			[-0.115,-0.042]	[-78.57,22.22]

Sign repair need not retain the other endpoints. Only 12/25/3/1 of the 18/30/5/2 repairs in B also retain the original screen and positive target contribution (Table 5). The joint feasible-oracle counts are 12/26/3/2. Small VOC repair subsets make their cost intervals descriptive, not evidence of stable population effects. Across *all baseline failures*, WF sign-repairs 42/48/11/6 cases; 0/4/0/0 opposite-sign regressions leave net reductions of 42/44/11/6. Across all 663/614/50/38 switches, 611/532/46/35 improve margin, with B/16 bbox changes of $-4.95/-13.20$ points on COCO/VOC. These are different populations from B repairs. Raw-target intervals containing zero do not establish equivalence.

Table 6: An illustrative COCO B/16 sign repair at $\epsilon = .02$ (image 254807, target: person). Selected by median baseline P among 17 repairs in $A^+ \cap F$; it does not represent repair frequency. Margins use each region’s own full foil set.

Endpoint	CCI-top1	WF
Worst-foil margin P	-.6545	+.2689
Mean-foil margin	+.0216	+.9090
Raw target drop	+.3462	+.3633
BBox precision	.7143	.6111

A sign repair can retain the screen but lose locality. Table 6 illustrates this distinction: the margin changes sign, raw target response remains positive, and bbox precision stays above .5, yet falls from .7143 to .6111. It is a response-level example; no unavailable spatial mask is reconstructed. Population-level retention is quantified by j in Table 5.

5 INTERPRETATION AND LIMITATIONS

The controls distinguish the effect of excluding annotated co-occurring classes from the effect of reducing foil count. Residual failures remain ontology- and screen-dependent; they do not establish complete annotation, semantic error, or a causal explanation of the response vector.

The audit concerns CCI-top1 on two datasets, two checkpoints, eight candidates, and one internal intervention. Region sizes are not matched and bbox precision does not measure object coverage or segmentation quality. Attention masking can change responses through other tokens, so spatial overlap alone cannot locate the causal source of a score change. Raw versus normalized worst-foil comparisons, nonzero failure thresholds, alternative candidate/intervention settings, and segmentation controls remain untested. Prompt holdout separates selection from evaluation templates, not models or images. Cross-checkpoint repair rates also depend on the uncalibrated target budget.

Reproducibility scope. Per-image candidate summaries, foil diagnostics, and analysis scripts support reanalysis of the reported endpoints. Public diagnostics omit full per-class response tensors and spatial masks; they do not support arbitrary new foil subsets or independent verification of region identity. This limitation is separate from the exact enumeration of the eight stored candidate responses.

6 CONCLUSION

Fixed-population foil controls show that annotated co-occurrence alone does not explain screened worst-foil failures under this protocol. The unrestricted oracle reveals lower candidate repair capacity on COCO than VOC; a tight target budget removes most remaining opportunities, while WF misses few feasible sign repairs. Relaxing the budget increases opportunity and costs. These results motivate reporting specificity, positive target contribution, and locality separately when auditing a selected explanation region.

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